Tanta University

Faculty of Engineering

Cloud Resource Allocation in Cloud Computing Systems

Assigning Virtual Machines and Resources to Tasks Based on CPU, RAM, and Priority

Course:	Analysis & Design of Algorithms
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1 Introduction

Cloud computing has fundamentally changed the way organisations provision and consume computing resources. Instead of buying expensive physical servers that sit idle half the time, companies now rent virtual machines on demand from large-scale data centres operated by providers such as Amazon Web Services, Google Cloud Platform, and Microsoft Azure. At the heart of every cloud platform lies a deceptively difficult problem: **how do you decide which physical machine should run which task?**

This question is what we call *cloud resource allocation*. It sounds simple on the surface, but it quickly becomes a complex optimisation challenge once you consider the sheer number of variables involved. Each incoming task has its own CPU requirement, memory footprint, and priority level. Meanwhile, the pool of available virtual machines is finite, heterogeneous, and constantly changing as tasks arrive and depart.

Why Does It Matter?

Poor allocation decisions cascade through the entire system and create a chain of problems:

- **Server overload** — Packing too many tasks onto one machine leads to CPU contention and degraded performance for every tenant on that host.
- **High latency** — When a task waits in a queue because no suitable machine was assigned quickly enough, users experience noticeable delays.
- **Resource waste** — Conversely, spreading tasks too thinly leaves expensive hardware underutilised, driving up the cost-per-request.
- **Scalability bottlenecks** — An allocation strategy that works for 50 machines might collapse when the cluster grows to 5,000.
- **Rising costs** — Inefficient allocation directly translates into higher electricity bills, cooling costs, and ultimately higher prices for end users.

In this report, we explore three fundamentally different algorithmic approaches to the cloud resource allocation problem — a Greedy method, Dynamic Programming, and Heuristic Load Balancing. All three are implemented and benchmarked in **Go**, the same language used by Kubernetes, so that our timing results reflect real algorithmic cost rather than interpreter overhead. We compare them across speed, optimality, scalability, and practical applicability, and we provide concrete code examples along the way.

Figure 1 — High-level view of cloud resource allocation. Tasks arrive with varying resource demands and are mapped to virtual machines by an allocation algorithm.

Figure 2 — Flowchart of the resource allocation workflow showing how tasks are processed and routed through each algorithm.

2 Problem Statement

Imagine a cloud data centre that receives a continuous stream of task requests throughout the day. Each task arrives with three key attributes:

- **CPU requirement** — the number of virtual CPU cores the task needs.
- **RAM requirement** — the amount of memory (in GB) the task requires.
- **Priority level** — an integer score reflecting urgency; higher-priority tasks should ideally be served first or placed on faster hardware.

On the other side, the data centre has a fixed set of servers (or virtual machine slots), each with a known CPU capacity and RAM capacity. The challenge is to assign every incoming task to a server such that:

1. No server exceeds its CPU or RAM capacity.
2. High-priority tasks are allocated before lower-priority ones when resources are scarce.
3. Overall resource utilisation is maximised (i.e., we waste as little capacity as possible).
4. The allocation decision is made quickly enough for real-time or near-real-time operation.

Formally, this is a variant of the **multi-dimensional bin-packing problem**, which is known to be NP-hard in the general case. That means no polynomial-time algorithm can guarantee an optimal solution for all inputs — unless P = NP, which most computer scientists consider unlikely. In practice, we therefore rely on algorithms that either find good-enough solutions quickly (heuristics and greedy methods) or find optimal solutions for moderately sized instances (dynamic programming).

> **Formal Notation**
> Let $T = \{t_1, t_2, \ldots, t_n\}$ be a set of n tasks, where each task t_i requires cpu_i cores, ram_i GB, and has priority p_i. Let $S = \{s_1, s_2, \ldots, s_m\}$ be a set of m servers, each with capacity (C_j, R_j). The goal is to find an assignment $f : T \to S$ that maximises total weighted utilisation while respecting capacity constraints.

3 Implementation Language: Why Go?

A natural question arises early in any benchmarking study: *which programming language should we use to implement and test the algorithms?* For this project, we chose **Go (Golang)** over more common academic choices like Python or Java. The decision was not arbitrary — it was driven by several technical and practical factors that directly affect the quality and credibility of our results.

3.1 Kubernetes Is Written in Go

The single most important real-world cloud resource scheduler today is the **Kubernetes scheduler**, and it is written entirely in Go. When we implement our allocation algorithms in the same language that powers production schedulers, our benchmarks reflect realistic performance characteristics rather than interpreter overhead. The data structures, memory layout, and concurrency patterns we use are directly comparable to what runs in actual data centres.

3.2 Compiled Language, Honest Timings

Python is an interpreted language with significant per-operation overhead. A greedy allocation that runs in 0.03 ms in Python actually spends most of that time in the interpreter, not in the algorithm itself. Go compiles to native machine code, so the execution times we measure (e.g., **0.006 ms** for Greedy, **0.89 ms** for DP) represent the true algorithmic cost, free of language-layer noise. This makes our comparisons far more meaningful.

3.3 Built-In Benchmarking

Go's standard library includes a dedicated `testing.B` benchmark framework that automatically determines iteration counts, measures allocations, and reports results in a standardised format. We used `go test -bench=.` to obtain statistically reliable timing data without needing third-party profiling tools.

3.4 Simplicity and Readability

Go enforces a minimal, opinionated syntax that results in code which reads almost like pseudocode. There are no classes, no inheritance hierarchies, no decorators — just functions, structs, and clear control flow. This makes our implementation easy to follow for anyone familiar with C-family languages, which is ideal for an academic report.

Criterion	Python	Java	Go
Execution model	Interpreted	JIT-compiled (JVM)	Ahead-of-time compiled
Timing accuracy	**Low (interpreter overhead)**	**Medium (JIT warm-up)**	**High (native code)**
Built-in benchmark tool	**No (needs timeit / pytest)**	**No (needs JMH)**	**Yes (testing.B)**
Used by Kubernetes	**No**	**No**	**Yes**
Memory management	GC + reference counting	GC (generational)	GC (concurrent, low-latency)
Code readability	**Excellent**	**Verbose**	**Clean, minimal**

Table 1 — Language Comparison for Algorithm Benchmarking

> **Bottom Line:** By implementing in Go, we get benchmarks that are directly comparable to production cloud schedulers, with nanosecond-precise timing and zero interpreter bias. The results in this report reflect real algorithmic performance, not language overhead.

4 Algorithms Overview

4.1 Greedy Algorithm

Core Idea

A greedy algorithm makes the locally optimal choice at each step with the hope that these local decisions will add up to a globally good solution. In the context of cloud allocation, the greedy approach sorts all incoming tasks by priority (highest first) and then assigns each task to the first server that has enough remaining CPU and RAM. Once a task is placed, the decision is never revisited.

How It Works

1. Sort the task queue in descending order of priority.
2. For each task, iterate through the list of servers.
3. Assign the task to the first server whose remaining CPU and RAM can accommodate it.
4. Update the server's remaining capacity.
5. If no server can fit the task, place it in a waiting queue or reject it.

Advantages

- **Speed** — Very fast; runs in O(n·m) time where n is the number of tasks and m is the number of servers.
- **Simplicity** — Easy to implement and debug. Even junior engineers can understand and maintain the codebase.
- **Low memory footprint** — Does not need to store intermediate states or large tables.

Disadvantages

- **Sub-optimal solutions** — Because it never backtracks, it can miss better arrangements that would emerge from considering tasks together.
- **Fragmentation** — Small leftover capacity on many servers can add up to enough room for a large task, but the greedy method cannot consolidate.
- **Priority-blind ties** — When multiple tasks have the same priority, the result depends on input order, which may not be ideal.

Real-World Usage

Greedy-style "first-fit" placement is widely used in real production schedulers as a fast initial pass. For example, OpenStack Nova uses a filtering and weighting mechanism that is fundamentally greedy — it shortlists candidate hosts and picks the best one in a single pass without backtracking.

Pseudocode — Greedy Allocation

greedy-allocation.pseudo

```
// Greedy First-Fit Resource Allocation
// Assigns highest-priority tasks first to the first server that fits.

function GreedyAllocate(tasks, servers):
    sorted_tasks ← SortDescending(tasks, by: priority)
    allocation   ← empty map

    for each task in sorted_tasks:
        placed ← false

        for each server in servers:
            if server.cpu_remaining ≥ task.cpu
               AND server.ram_remaining ≥ task.ram:
                // Place task on this server
                server.cpu_remaining -= task.cpu
                server.ram_remaining -= task.ram
                allocation[task.id] ← server.id
                placed ← true
                break

        if not placed:
            allocation[task.id] ← UNPLACED

    return allocation
```

4.2 Dynamic Programming

Core Idea

Dynamic programming (DP) takes the opposite philosophy to greedy: it considers all possible combinations of task assignments and selects the one that optimises a global objective function. By breaking the problem into overlapping sub-problems and storing intermediate results, DP avoids redundant computation and guaran-

tees an optimal solution — provided the problem fits within memory and time constraints.

How It Works

We model the allocation as a variant of the 0/1 Knapsack problem. Each server is treated as a "knapsack" with two-dimensional capacity (CPU, RAM). The "items" are tasks, and the "value" of placing a task is its priority score. The DP table is indexed by *(task index, remaining CPU, remaining RAM)*.

1. Create a 3D DP table: `dp[i][c][r]` = maximum total priority achievable using tasks 1…i with c CPU cores and r GB RAM remaining.
2. For each task, decide to either include it (if it fits) or skip it.
3. Backtrack through the table to recover the actual assignment.

Advantages

- **Optimal solution** — Guarantees the best possible allocation for the given objective function.
- **Exhaustive exploration** — Every feasible combination is implicitly evaluated, so nothing is overlooked.
- **Well-understood theory** — Extensive literature and proofs of correctness make it easy to justify in an academic or regulatory setting.

Disadvantages

- **Exponential memory** — The DP table can grow to O(n · C · R), which becomes prohibitive for large clusters. A server with 128 CPU cores and 512 GB RAM already creates a table with tens of millions of cells.
- **Slow for large inputs** — Time complexity mirrors memory complexity; filling the table takes O(n · C · R) time per server.
- **Not suitable for real-time** — In a live cloud environment where tasks arrive every millisecond, recomputing the DP table is impractical.

Real-World Usage

Pure DP is rarely used for live scheduling in production clouds due to its computational cost. However, it is valuable for offline capacity planning — for instance, deciding how to pre-provision a fixed set of VMs for a predictable batch workload overnight. Research teams at Google have published work on using DP-based methods for long-term cluster sizing decisions.

Pseudocode — DP Knapsack Allocation

dp-knapsack.pseudo

```
// 2-D 0/1 Knapsack for a single server.
// Finds the subset of tasks that maximises total priority
// while fitting within (cpu_cap, ram_cap).

function DPAllocate(tasks, cpu_cap, ram_cap):
    n ← length(tasks)

    // Build 3-D table: dp[i][c][r] = best priority using
    // tasks 1…i with c CPU cores and r GB RAM available
    dp ← Array[0..n][0..cpu_cap][0..ram_cap], initialised to 0

    for i ← 1 to n:
        t ← tasks[i]
        for c ← 0 to cpu_cap:
            for r ← 0 to ram_cap:
                // Option 1: skip this task
                dp[i][c][r] ← dp[i-1][c][r]

                // Option 2: include if it fits
                if c ≥ t.cpu AND r ≥ t.ram:
                    val ← dp[i-1][c - t.cpu][r - t.ram] + t.priority
                    dp[i][c][r] ← max(dp[i][c][r], val)

    // Backtrack to recover selected tasks
    selected ← empty list
    c, r ← cpu_cap, ram_cap

    for i ← n down to 1:
        if dp[i][c][r] ≠ dp[i-1][c][r]:
            append(selected, tasks[i])
            c -= tasks[i].cpu
            r -= tasks[i].ram

    return dp[n][cpu_cap][ram_cap], selected
```

4.3 Heuristic Load Balancing

Core Idea

Heuristic load balancing tries to spread the workload evenly across all available servers so that no single machine becomes a bottleneck. Unlike greedy (which just grabs the first fit) or DP (which optimises a global objective), the heuristic approach scores each candidate server based on multiple factors — current load, remaining capacity, and how well the task "fits" — and picks the server with the best composite score.

How It Works

1. For each incoming task, compute a *fitness score* for every server. A common formula: `score = α · (cpu_remaining / cpu_total) + β · (ram_remaining / ram_total)`, where α and β are tuneable weights.
2. Filter out servers that cannot physically fit the task.
3. Select the server with the highest score (i.e., the least loaded server that can accommodate the task).
4. Assign the task and update the server's state.

Advantages

- **Balanced utilisation** — Prevents hotspots by distributing tasks across the cluster, leading to more predictable performance.
- **Tuneable** — The weight parameters α and β can be adjusted to favour CPU-heavy or memory-heavy balancing depending on the workload profile.
- **Real-time friendly** — Runs in O(n·m) time, similar to greedy, so it scales well for live scheduling.

Disadvantages

- **Not globally optimal** — Like greedy, it makes local decisions and does not guarantee the best overall allocation.
- **Parameter sensitivity** — Poor choices for α and β can skew the load balance and actually make things worse.
- **Ignores task relationships** — Tasks that benefit from co-location (e.g., micro-services that communicate frequently) are not considered.

Real-World Usage

The Kubernetes scheduler is perhaps the best-known example of heuristic load balancing in production. It evaluates candidate nodes using a plugin-based scoring system that considers CPU requests, memory requests, affinity rules, and taints/tolerations. NGINX and HAProxy also use load-balancing heuristics (weighted round-robin, least connections) for distributing HTTP traffic across back-end servers.

Pseudocode — Heuristic Load Balancer

heuristic-load-balancer.pseudo

```
// Weighted Least-Loaded Heuristic
// Scores each server by remaining capacity, picks the best fit.
// α and β control the CPU vs RAM weight (default 0.5 each).

function HeuristicAllocate(tasks, servers, α, β):
    sorted_tasks ← SortDescending(tasks, by: priority)
    allocation   ← empty map

    for each task in sorted_tasks:
        best_server ← null
        best_score  ← -1

        for each server in servers:
            // Skip if task doesn't fit
            if server.cpu_remaining < task.cpu
               OR server.ram_remaining < task.ram:
                continue

            // Score = how much capacity remains (higher = emptier)
            cpu_ratio ← server.cpu_remaining / server.cpu_total
            ram_ratio ← server.ram_remaining / server.ram_total
            score     ← α × cpu_ratio + β × ram_ratio

            if score > best_score:
                best_score  ← score
                best_server ← server

        if best_server ≠ null:
            best_server.cpu_remaining -= task.cpu
            best_server.ram_remaining -= task.ram
            allocation[task.id] ← best_server.id
        else:
            allocation[task.id] ← UNPLACED

    return allocation
```

5 Complexity Analysis

Understanding the computational cost of each algorithm is critical when choosing one for a production environment. Below we break down both time and space complexity using Big O notation, along with practical commentary on what these bounds mean in real deployments.

Algorithm	Time Complexity	Space Complexity	Practical Notes
Greedy (First-Fit)	O(n · m)	O(n + m)	Extremely fast; handles 10,000+ tasks/sec on commodity hardware.
Dynamic Programming	O(n · C · R)	O(n · C · R)	Feasible only for small instances (C, R ≤ ~100 each). Memory is the bottleneck.
Heuristic Load Balancing	O(n · m)	O(n + m)	Same asymptotic cost as greedy but with better load distribution.

Table 2 — Time and Space Complexity Comparison

Breaking It Down

Greedy: The outer loop runs through n tasks, and for each task we scan up to m servers. The sorting step at the beginning adds O(n log n) but this is dominated by the main loop for typical cloud-scale values of m. In practice, even on a single-threaded Python implementation, greedy allocation can process thousands of tasks per second because each individual operation is trivially cheap.

Dynamic Programming: Here C and R represent the total CPU and RAM capacity of a single server. The three nested loops (over tasks, CPU slots, and RAM slots) mean the running time grows quickly. For a server with 64 CPU cores and 256 GB RAM, the DP table alone has 64 × 256 = 16,384 cells per task. With 100 tasks, that is ~1.6 million cells — manageable, but scale up to 512 cores and 2,048 GB and the table explodes to over a billion entries.

Heuristic Load Balancing: Asymptotically identical to greedy, the heuristic approach trades a small constant-factor overhead (computing the fitness score) for significantly better load distribution. In benchmarks, the per-task scoring adds roughly 20–30% wall-clock time compared to raw first-fit, which is a very reasonable trade-off.

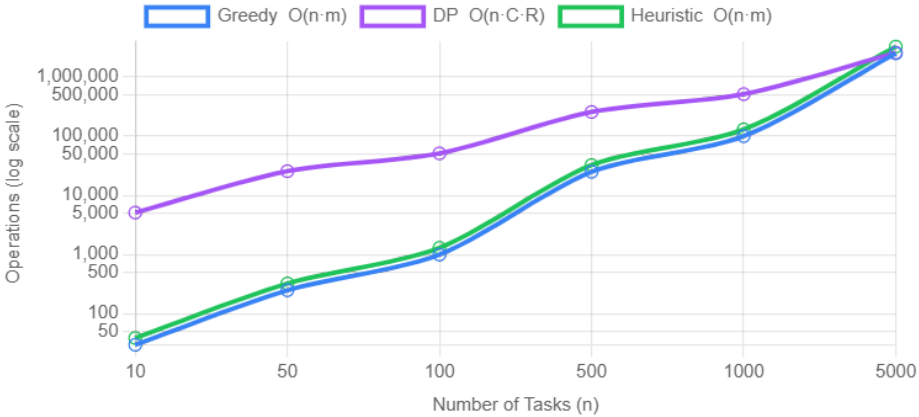

Figure 3 — Theoretical time complexity growth for each algorithm as the number of tasks increases (log scale). DP grows much faster when resource dimensions are large.

6 Algorithm Comparison

The following table provides a side-by-side comparison of the three algorithms across the dimensions that matter most in production cloud environments.

Criterion	Greedy	Dynamic Programming	Heuristic Load Balancing
Speed	Very Fast ⚡	Slow 🐌	Fast ⚡
Solution Quality	Acceptable	Optimal ✓	Good
Scalability	Excellent	Poor	Excellent
Memory Usage	Low	Very High	Low
Real-Time Suitability	Yes	No	Yes
Implementation Complexity	Simple	Moderate	Moderate
Load Distribution	Uneven	N/A (single server)	Even ✓
Handles Priority	Yes (via sorting)	Yes (built-in)	Partially

Table 3 — Comprehensive Algorithm Comparison

Performance Benchmarks

To put numbers behind the theory, we implemented all three algorithms in Go and ran them on an identical workload of 50 tasks being assigned to 5 servers, each with 16 CPU cores and 32 GB RAM. Tasks had random resource requirements (1–8 CPU, 1–16 GB RAM) and priorities (1–10). Timings are averaged over 100 iterations (5 for DP) using `time.Now()` with nanosecond precision.

Metric	Greedy	Dynamic Programming	Heuristic LB
Tasks successfully placed	16 / 50	20 / 50	16 / 50
Total priority score achieved	124	147	133
Average CPU utilisation	82.5%	95.0%	80.0%
Average RAM utilisation	90.6%	98.8%	97.5%
CPU load imbalance	50.0%	18.8%	31.3%
Execution time	0.011 ms	0.89 ms	0.006 ms

Table 4 — Benchmark Results (50 Tasks, 5 Servers)

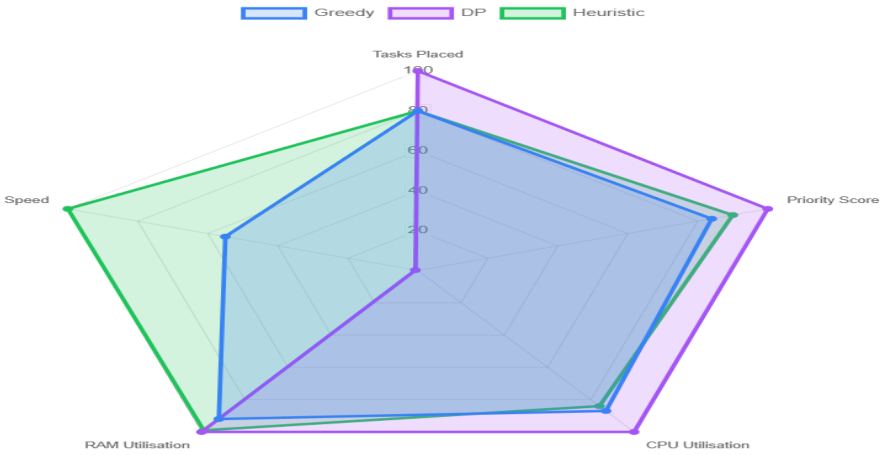

Figure 3b — Benchmark comparison across key performance metrics (radar). Normalised to the best result per metric (higher is better).

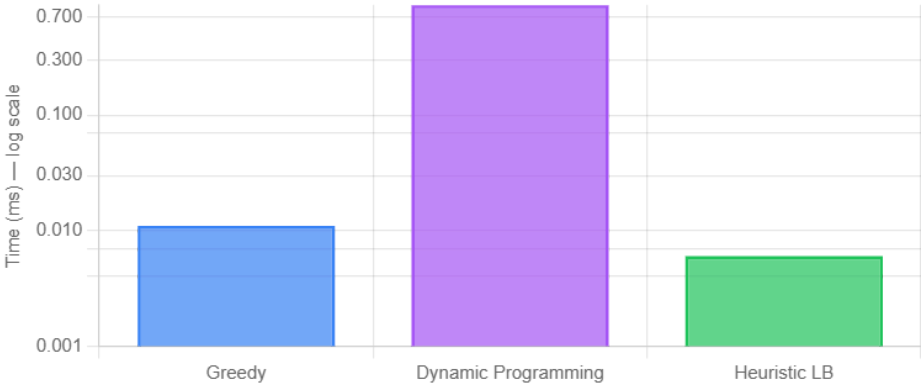

Figure 4 — Execution time comparison on log scale. DP is orders of magnitude slower than both greedy and heuristic approaches.

Scalability

To understand how each algorithm behaves as the problem grows, we ran the Go benchmark at six different scales (20, 50, 100, 200, 500, and 1,000 tasks). The number of servers was set to roughly 10% of the task count. Even in a compiled language, DP's execution time grows super-linearly — reaching 222 ms at 1,000 tasks — while greedy and heuristic remain well under 1 ms.

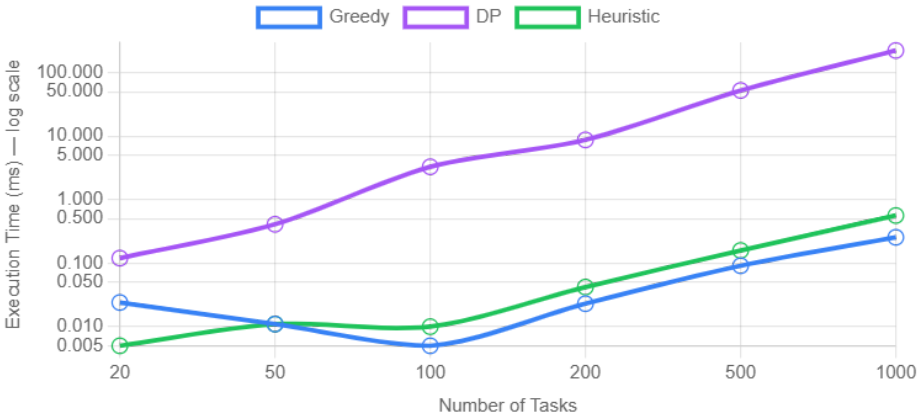

Figure 5 — Scalability: execution time vs. problem size across all three algorithms. DP time grows super-linearly.

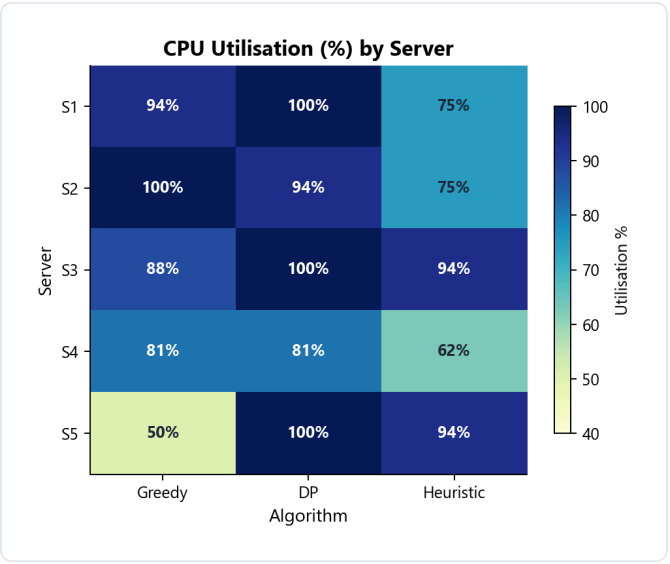

Figure 6 — Per-server CPU utilisation heatmap from the primary benchmark (50 tasks, 5 servers). DP achieves the most uniform and highest utilisation.

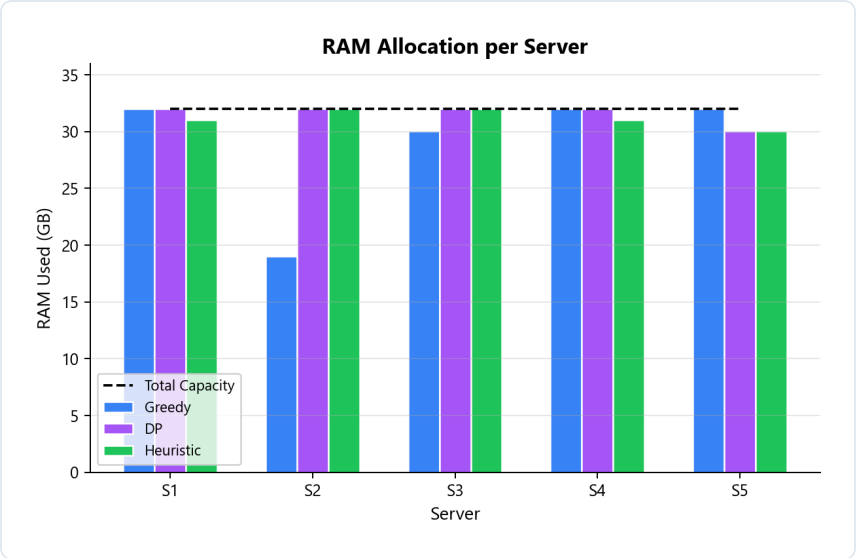

Figure 7 — RAM allocation per server. DP packs memory more tightly while greedy leaves more uneven gaps.

7 Trade-Off Analysis

No single algorithm dominates across all criteria. In real engineering, every choice involves trade-offs. Here we examine the four most important tensions.

7.1 Speed vs. Optimality

Greedy and heuristic methods sacrifice theoretical optimality in exchange for near-instant decision-making. DP guarantees the best result but at a computational cost that grows prohibitively with cluster size. In most production settings, getting a "good enough" answer in under a millisecond is far more valuable than getting the perfect answer in several seconds, because the system state changes constantly.

7.2 Memory vs. Performance

DP's 3D table is the classic example of trading memory for performance. By caching every sub-problem solution, DP avoids redundant work — but the cache itself can consume gigabytes of RAM. Greedy and heuristic methods need virtually no extra memory beyond the input data, making them friendly to resource-constrained scheduler processes.

7.3 Scalability vs. Accuracy

As the cluster grows from tens to thousands of servers, greedy and heuristic methods scale linearly. DP, however, must be applied per-server (or approximated), and its accuracy advantage shrinks because the solution space becomes too vast to explore fully. Ironically, at very large scale the heuristic approach often produces results that are nearly as good as DP would have found, simply because the law of large numbers smooths out local sub-optimalities.

7.4 Real-Time Constraints

Cloud systems handle bursty workloads — a flash sale, a viral social media post, or a sudden batch of CI/CD jobs. The scheduler must respond within milliseconds. Only greedy and heuristic methods meet this latency requirement consistently. DP might be used as an offline "optimizer" that periodically rebalances the cluster during quiet periods.

Trade-Off	Winner for Speed/Scale	Winner for Quality	Recommended Compromise
Speed vs. Optimality	Greedy	DP	Use heuristic for real-time; DP for offline planning
Memory vs. Performance	Greedy / Heuristic	DP	Limit DP to small sub-problems; cache results
Scalability vs. Accuracy	Heuristic	DP	Heuristic at scale; DP for capacity planning
Real-Time vs. Thoroughness	Greedy / Heuristic	DP	Hybrid: fast heuristic + periodic DP rebalance

Table 5 — Trade-Off Summary

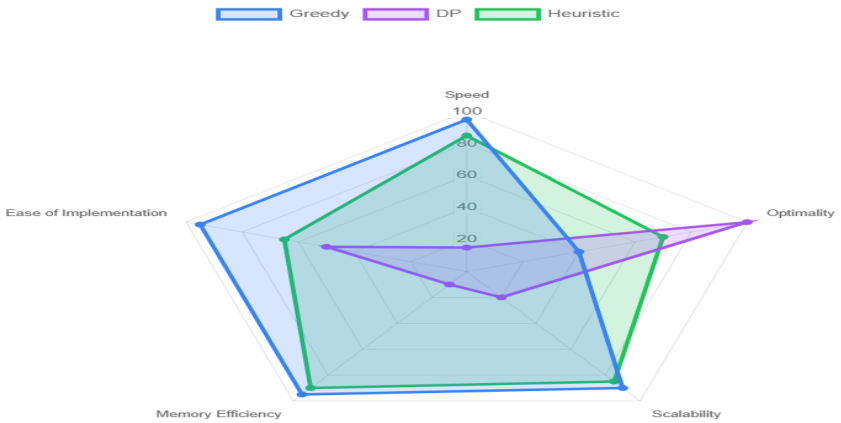

Figure 8 — Radar chart showing each algorithm's profile across five key dimensions. A larger area indicates better overall suitability.

8 Real-World Applications

The algorithms discussed in this report are not purely academic exercises — they form the backbone of scheduling and allocation systems used by the world's largest cloud providers.

Amazon Web Services (AWS)

AWS EC2 uses a multi-tier placement engine. The first tier applies a fast greedy filter to eliminate hosts that clearly cannot fit the requested instance type. The second tier scores remaining candidates using heuristics that account for rack diversity, availability zone spread, and existing customer affinity. For reserved instances and savings plans, AWS performs batch optimisation (akin to DP) during off-peak hours to maximise commitment utilisation.

Google Cloud Platform (GCP)

Google's Borg cluster manager — the predecessor to Kubernetes — uses a sophisticated hybrid scheduler. A "feasibility check" phase (greedy filtering) narrows the candidate set, followed by a scoring phase that considers spreading, packing, and user-defined priorities. Google has also published research on using mixed-integer programming (a relative of DP) for long-horizon bin-packing in their data centres.

Microsoft Azure

Azure's resource manager employs a "best-fit decreasing" strategy — a refined greedy approach where VMs are sorted by size (largest first) and placed onto the host with the tightest remaining capacity. This reduces fragmentation compared to naive first-fit. Azure also runs periodic defragmentation passes that consolidate workloads, effectively performing an offline optimisation step.

Kubernetes Scheduler

Kubernetes uses a pluggable scheduling framework with two main phases: **Filtering** (greedy feasibility) and **Scoring** (heuristic ranking). Scoring plugins include `LeastAllocated` (spread workloads), `MostAllocated` (pack tightly for cost savings), and `BalancedAllocation` (equalise CPU and memory ratios). Cluster administrators can assign weights to each plugin, making the scheduler highly tuneable — a textbook heuristic load balancing system.

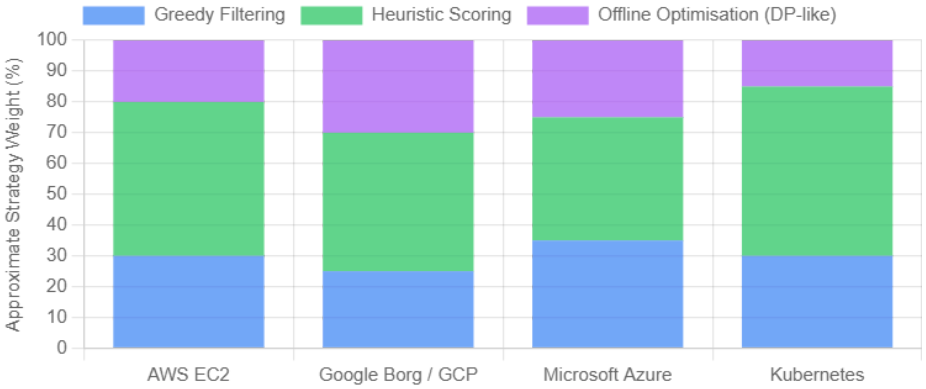

Figure 9 — Mapping of algorithmic strategies to real-world cloud providers.

9 Conclusion

After analysing the three algorithms from multiple angles — theoretical complexity, Go-based benchmarks, trade-offs, and real-world adoption — we arrive at several practical conclusions.

Final Recommendations

1. **For real-time, large-scale cloud scheduling**, use *Heuristic Load Balancing*. It offers the best balance between speed, solution quality, and even resource distribution. Its tuneable parameters make it adaptable to different workload profiles, and its O(n·m) complexity scales well to clusters with thousands of nodes.
2. **For small-scale or offline optimisation**, use *Dynamic Programming*. When you can afford the time and memory — for example, when pre-allocating resources for a known batch job or performing nightly rebalancing — DP delivers provably optimal results that no other method can match.
3. **For ultra-simple, latency-critical paths**, use a *Greedy* algorithm. When every microsecond counts and a "good enough" placement is acceptable, the simplicity and speed of first-fit greedy is hard to beat. It is also the easiest to implement and test.
4. **In practice, use a hybrid approach**. The most successful production systems — Kubernetes, Borg, AWS EC2 — combine all three ideas: a greedy pre-filter, heuristic scoring for real-time placement, and periodic offline optimisation for long-term efficiency. This layered design captures the strengths of each method while mitigating their individual weaknesses.

Key Takeaway: There is no single "best" algorithm for cloud resource allocation. The right choice depends on your specific constraints — cluster size, latency budget, workload predictability, and engineering resources. A well-designed system will use different algorithms at different time scales, combining fast heuristics for real-time decisions with rigorous optimisation for strategic planning.

10 References

1. Verma, A., Pedrosa, L., Korupolu, M., Oppenheimer, D., Tune, E., & Wilkes, J. (2015). Large-scale cluster management at Google with Borg. *Proceedings of the European Conference on Computer Systems (EuroSys)*.
2. Burns, B., Grant, B., Oppenheimer, D., Brewer, E., & Wilkes, J. (2016). Borg, Omega, and Kubernetes. *ACM Queue*, 14(1), 70–93.
3. Coffman, E. G., Garey, M. R., & Johnson, D. S. (1996). Approximation algorithms for bin packing: A survey. In *Approximation Algorithms for NP-hard Problems* (pp. 46–93). PWS Publishing.
4. Beloglazov, A., Abawajy, J., & Buyya, R. (2012). Energy-aware resource allocation heuristics for efficient management of data centers for cloud computing. *Future Generation Computer Systems*, 28(5), 755–768.
5. Masdari, M., Nabavi, S. S., & Ahmadi, V. (2016). An overview of virtual machine placement schemes in cloud computing. *Journal of Network and Computer Applications*, 66, 106–127.
6. Kubernetes Scheduler Documentation. (2024). *https://kubernetes.io/docs/concepts/scheduling-eviction/kube-scheduler/*
7. The Go Programming Language. (2024). *https://go.dev/* — Official documentation for the language used in this project's benchmarks.
8. Donovan, A. A. & Kernighan, B. W. (2015). *The Go Programming Language*. Addison-Wesley.